The $(\log 2, \log 3]$ step:

Connes–Consani's archimedean operator rebuilt, the semi-local $\{\infty, 2\}$ Sonin space on a slice,
and why the compact-remainder mechanism does not cross the place 2

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3 September 2026, v4 (5 September: 2-adic weight open again after the $\Lambda \geq 16$ campaign) — consolidates notebook §80–85, §88 of `riemann-weil-phenomenology`

Abstract

Weil positivity is known, without RH, for test functions supported in $[2^{-1/2}, 2^{1/2}]$ (window length $L = \log 2$); the proof of Connes and Consani (Selecta 2021) writes the archimedean Weil functional as a positive trace on Sonin's space minus a remainder whose conditioned operator is $-2\varepsilon'(1^+)\,\mathrm{Id}$ plus a compact part with a single eigenvalue above 1. Crossing the threshold of the prime 2 — positivity for $L \in (\log 2, \log 3]$ — is the first open step of their semi-local program. We (i) rebuild their archimedean operator and reproduce every published digit; (ii) measure that its compact part acquires a second eigenvalue above 1 at $L \approx 1.01$, while the archimedean form itself stays positive on the conditioned hyperplane at $L = \log 3$; (iii) show that the prime-2 term cannot be added to the compact remainder; (iv) build the semi-local $\{\infty, 2\}$ Sonin space on the $\mathrm{ord}_2 = 0$ slice, where the adelic Fourier transform has the closed form $\mathfrak{F}g(\rho) = \frac{1}{2}[\sum_{n\geq 0} \hat{g}(2^n\rho) - \hat{g}(\rho/2)]$, and find that the compression is not trace class, that the remainder is log-divergent at $\rho = 1$ and spiked at $\rho = 2$, and that the conditioned remainder, tested on twenty smooth functions at two window lengths, is predominantly positive (20/20 at $\log 2$, 13/20 at $\log 3$) where the archimedean one is essentially negative — the sign that made the archimedean argument work is reversed on every test function tried; (v) check the construction against the semi-local trace formula of Connes (1999): the archimedean distribution is reproduced to 1–4% and the 2-adic place appears exactly at $\lambda = 2^{\pm 1}$; the mass of the peak is not yet settled — it converges toward $(\log 2)/\sqrt{2}$ by cell extrapolation at $\Lambda = 4$ but climbs through and past that value at $\Lambda = 16$ as the cells are refined. Everything is measurement and calibration; the conclusion is that the $(\log 2, \log 3]$ step requires a mechanism of a different nature, and the construction points to the term in the way. Nothing here bears on RH.

1 The archimedean mechanism and where it stops

Connes–Consani prove, for g supported in $[2^{-1/2}, 2^{1/2}]$ with $\hat{g}(\pm i/2) = \hat{g}(0) = 0$, that $W_\infty(g * g^*) \geq \mathrm{Tr}(\vartheta(g)S\vartheta(g)^*) \geq 0$. The engine is the identity (their Theorem 4.7) $\mathrm{Tr}(\vartheta(f)S) = W_\infty(f) + E(f)$, S the projection on Sonin's space $\mathcal{S}(1,1)$ (even functions vanishing with their Fourier transform on $[-1,1]$), $E(f) = \int f(\rho^{-1})\epsilon(\rho)\, d^*\rho$ with ϵ explicit in the prolate spheroidal functions of bandwidth 2π. After the vanishing conditions are imposed through $Q = -(\rho\partial_\rho)^2 + \frac{1}{4}$, the remainder becomes $E \circ Q(\xi * \xi^*) = \langle \xi | N_I \xi \rangle$, $N_I = -2\epsilon'(1^+)(\mathrm{Id} - K_I)$, K_I compact of Hilbert–Schmidt class on $L^2(I)$, $I = [-\frac{1}{2}\log 2, \frac{1}{2}\log 2]$. Positivity follows if $K_I \leq 1$; they compute that K_I has exactly one eigenvalue above 1 (1.0516), neutralized by one linear condition, with constant $c \in (13, 17)$. The kink of ϵ at $\rho = 1$ ($\epsilon'(1^+) = 22.9965$) is what produces the $-2\epsilon'\mathrm{Id}$; the finiteness $\delta(1) = \sum \lambda(n)^2 = 2.2375$ is what makes the remainder a regular kernel. The whole argument stops at $L = \log 2$, to one eigenvalue.

2 Reconstruction and calibration

`code/cc_arch.py` rebuilds the objects: prolates $PS_{2n,0}(2\pi,\cdot)$, eigenvalues $\lambda(n)$ of the finite Fourier transform, ξ_n, the analytic continuation $\xi_n^{an}(x) = \lambda(n)^{-1}\int_{-1}^{1}\xi_n(t)\cos(2\pi tx)\,dt$, $Q\epsilon$ from their formula (99), and K_I by their $q \to 1$ Toeplitz discretization ($\omega = 2 \times 10^{-3}$).

quantity	rebuilt	Connes–Consani
$\lambda(0), \lambda(1), \lambda(2), \lambda(3)$	0.999971, −0.979485, 0.524086, −0.058977	0.999971, −0.979485, 0.524086, −0.0589766
$t(0), t(1), t(2), t(3)$	11.9719, 8.77574, 2.20528, 0.0434	11.9719, 8.77574, 2.20528, 0.0433983
$\epsilon'(1^+)$	22.9965	22.9965
$\lambda_{\max}(K_I), \lambda_2, \lambda_3$ at $\log 2$	1.05176, 0.68791, 0.0297	1.05177 ($\omega = 10^{-3}$), 0.687925, 0.0289

3 Beyond $\log 2$ in the archimedean frame

The second eigenvalue of K_I crosses 1 at $L \approx 1.01$ ($\mu \approx 2.74$): one eigenvalue above 1 on $(\log 2, 1.01)$, two on $(1.01, \log 3]$ (1.0899, 1.0395 at $\log 3$). Yet the archimedean form itself is still positive in their frame: on the Galerkin space V_{31} at $\mu = 3$, the archimedean form with the pole removed has one negative direction on V (along the pole) and *none* on the hyperplane $c(f) = 0$ of the vanishing condition. Their sufficient condition $K_I \le 1$ thus already fails at $\log 3$ while the conclusion holds — the Sonin trace compensates both directions of the remainder.

The prime-2 term cannot be added to the remainder. On the conditioned functions ξ, $W_2 \circ Q(\xi * \xi^*) = \frac{\log 2}{\sqrt{2}}\big[\frac{1}{4}F(\log 2) - F''(\log 2)\big]$, $F = \xi * \xi^*$, is a differential form, unbounded on $L^2(I)$; on the side $g = (\frac{1}{2} - \partial)\xi$ it is a bounded translation form, but the archimedean identity term becomes $-2\epsilon'\|Y * g\|^2$, a weak norm that does not control $G(\log 2)$ for oscillating g. In either presentation $E + W_2 \le 0$ is false, although $W_\infty - W_2 \ge 0$ holds numerically at $\log 3$ ($\lambda_{\min} = 5.9 \times 10^{-8}$ on V_{31}): the positivity comes from the positive trace absorbing W_2, not from a bound on the remainder. This is the operator form of a Galerkin measurement: at $L = \log 3$ the archimedean form Q_∞ = pole + arch has exactly one negative direction v (eigenvalue -0.074), $-T_2$ is positive on it with margin ($Q(v) = +0.050$, using 5% of T_2's mass), and the razor 5.9×10^{-8} lives in the positive complement.

4 The semi-local $\{\infty, 2\}$ Sonin space on a slice

Following Connes (1999, §VII), $X_S = (\mathbb{R} \times \mathbb{Q}_2)/\pm 2^{\mathbb{Z}}$, $L^2(X_S)$ is the completion of $\mathcal{S}(\mathbb{A}_S)$ for $\|f\|^2 = \int_{\mathcal{D}} |\sum_q f(qx)|^2 |x|\, d^*x$, and $F = F_\infty \otimes F_2$ is unitary on it (his Lemma 1). Functions invariant under $\mathbb{Z}_2^*$ are determined by their restriction g to the shell $\mathrm{ord}_2 = 0$, and on this slice

$$\mathfrak{F}g(\rho) = \tfrac{1}{2}\Big[\sum_{n\ge 0}\hat{g}(2^n\rho) - \hat{g}(\rho/2)\Big], \qquad \hat{g}(\xi) = 2\int_0^\infty g(r)\cos(2\pi r\xi)\,dr. \tag{1}$$

Proposition 1. *$\mathfrak{F}$ is unitary and involutive on $L^2([0,\infty))$; hence self-adjoint.*

Proof. A dyadic Plancherel argument. Write $\mathfrak{F}g = \sum_{k\ge -1} c_k\,\hat{g}(2^k\cdot)$ with $c_{-1} = -\frac{1}{2}$, $c_k = \frac{1}{2}$ ($k \ge 0$), and expand $\|\mathfrak{F}g\|^2 = \sum_{k,l} c_k c_l \langle \hat{g}(2^k\cdot), \hat{g}(2^l\cdot)\rangle$. Group the pairs by $j = l - k$: $\langle \hat{g}(2^k\cdot), \hat{g}(2^{k+j}\cdot)\rangle = 2^{-k} I_j$ with $I_j = \langle \hat{g}, \hat{g}(2^j\cdot)\rangle$ independent of k, so the coefficient of I_j is $\sum_k c_k c_{k+j} 2^{-k}$, which equals $2 \cdot (-\frac{1}{4}) + \frac{1}{4}\sum_{k\ge 0} 2^{-k} = -\frac{1}{2} + \frac{1}{2} = 0$ for every $j \ne 0$, while for $j = 0$ it is $\frac{1}{4}\cdot 2 + \frac{1}{4}\cdot 2 = 1$. Hence $\|\mathfrak{F}g\|^2 = I_0 = \|\hat{g}\|^2 = \|g\|^2$. Involutivity: F^2 is the reflection $x \mapsto -x$ on $\mathcal{S}(\mathbb{A}_S)$, the identity on the even (i.e. ± 1-invariant) functions, and it descends to $L^2(X_S)$ because F commutes with the $\mathcal{O}_S^*$-action (Connes 1999, Lemma 1). □

Remark. The proposition is the analytic statement; the numbers $\|\mathfrak{F}g\|^2/\|g\|^2 = 1.015$ and $\mathfrak{F}^2 g = 0.94\, g$ on the support of a Gaussian bump (`code/semilocal.py`, tests) are a quadrature check of the implementation, with the truncation at $\rho \leq 130$ and the grid 10^{-3} as error sources; they are not part of the proof.

The cutoff $P_1 = \mathbf{1}_{[0,1]}$ and the scaling $\vartheta(\lambda)g(r) = \lambda^{-1/2}g(r/\lambda)$ are unchanged; only $\hat{P}_1 = \mathfrak{F}P_1\mathfrak{F}$ and Sonin's space change. Three facts, all computed with exact cell averaging (a first pointwise assembly aliased the lacunary terms and produced a spurious 43% asymmetry; it is retracted in the notebook):

- *The compression is not trace class.* The eigenvalues of $P_1\mathfrak{F}P_1$ decay slowly $(0.998, -0.990, 0.613, -0.554, 0.4$ and $\sum \lambda_n^2$ grows with resolution (4.24 at $N = 200$, 4.75 at $N = 400$; archimedean: 2.2375, stable). The kernel $\sum_n \cos(2\pi 2^n \rho r)$ is not square integrable on $[0,1]^2$.
- *The remainder is log-divergent at $\rho = 1$ and spiked at $\rho = 2$.* $\delta_S(\rho) = \mathrm{Tr}(\vartheta(\rho^{-1})\hat{P}_1 P_1) = \sum_n \lambda_n \langle \xi_n | \vartheta(\rho^{-1})\eta_n\rangle$ (CC's Lemma 4.1, which uses only the algebra of two projections). Archimedean control: the closed form $2\sqrt{\rho}\,[\mathrm{Si}(2\pi(1+\rho))/(2\pi(1+\rho)) + \mathrm{Si}(2\pi(\rho-1))/(2\pi(\rho-1))]$ is reproduced to 3–4 digits. Semi-local: $\delta_S \approx -0.65\ln(\rho-1) + C$ ($\delta_S(1) = \sum \lambda_n^2 = \infty$) — a logarithm where CC had a kink — and a spike at $\rho = 2$ growing with resolution: the 2-adic Weil term, appearing where it must.
- *The conditioned remainder is essentially positive.* On twenty smooth test functions of I (sine basis; eigenvalues of $D \circ Q$ relative to the Gram matrix; `code/dq_sign.py`): archimedean at $\log 2$, 2 positive eigenvalues out of 20 (4.77, 1.23) and the rest massed at -2.01 — the $-2\,\mathrm{Id}$ of CC's Theorem 3.6 in plain sight, and their Remark 3.9 recovered; at $\log 3$, 3 positive. Semi-local at $\log 2$: 20 **positive out of** 20 (up to 121); at $\log 3$, 13 out of 20 (up to 196), the positive part growing with the basis — the $H^{1/2}$ signature of the logarithm ($-\ln|x|$ and the finite part of $1/x^2$ both have positive transforms).

5 The verdict

$W_\infty - W_2 = L_S - D_S$ with L_S a positive trace. Extracting positivity requires $D_S \circ Q$ essentially negative, as in the archimedean case. It is essentially positive: the positive trace absorbs the place 2 but with a divergent positive surplus that erases the sign of W. The compact-remainder mechanism of Connes–Consani, transported verbatim to $\{\infty, 2\}$ at $\Lambda = 1$, does not give the $(\log 2, \log 3]$ step (28th preregistered execution of the repository). The construction also suggests why, as a reading of the semi-local trace formula rather than an identification of operators: the divergent surplus has the form of the term $2h(1)\log'\Lambda$ of Connes (1999), and that theorem states that the finite part of the compressed trace, once this term is subtracted, *is* the Weil functional — no positivity comes for free. The archimedean miracle (a finite $\delta(1)$ and a kink) belongs to one place.

6 Validation against the semi-local trace formula

$\tau_\Lambda(\lambda) = \mathrm{Tr}(\hat{P}_\Lambda P_\Lambda \vartheta(\lambda))$ on the slice (`code/trace_dist.py`, $\Lambda = 4, 8$). Theorem 4 of Connes (1999) predicts $\tau_\Lambda \to 2\log'\Lambda\,\delta_1 + \tau_\infty + \tau_2$ with $\tau_\infty = \lambda^{1/2}/2\,(1/(1+\lambda) + 1/|1-\lambda|)$ (CC (39)) and τ_2 point masses at $\lambda = 2^{\pm m}$. Archimedean: τ_Λ matches (39) to 1–4% away from $\lambda = 1$ (0.7495/0.7529 at 0.4, 2.4986/2.4845 at 0.8, 0.9338/0.9428 at 2.0, 0.6521/0.6495 at 3.0). Semi-local minus archimedean: two sharp peaks exactly at $\lambda = 2$ and $\lambda = \frac{1}{2}$ at both Λ — the 2-adic place where the theorem puts it. Their integrated weights depend on the cell size, not on Λ: the peaks are narrower than a cell ($\sim \Lambda^{-2}$ against $h = 1/32$), and a first pass at fixed cells read aliased, sign-changing masses ($+0.13$, -0.23). Refining the cells at fixed $\Lambda = 4$ gives $0.17 \to 0.32$ at $\lambda = 2$ for $h = 1/32 \to 1/64$, and the $h \to 0$ extrapolation of the note *visibility-offline* (Grok, 3 September) reaches 0.488 and 0.491, the $k = 1$ term of the local Weil factor $(\log 2)/\sqrt{2} = 0.4901$. But the campaign at $\Lambda \geq 16$ (Grok, 4–5 September, `report/campaign_2adic_large.csv`) finds the mass at $\lambda = 2$ still climbing with resolution — $0.14, 0.26, 0.35, 0.44, 0.59$ for $h = 1/80 \ldots 1/160$ at $\Lambda = 16$, 0.57 at $\Lambda = 24$ — through and past 0.49, a Hann taper changing nothing, and the mass at $\lambda = \frac{1}{2}$ far above it. The check is conclusive on the location of the places; the 2-adic weight is *open*: the two extrapolations disagree, the peak width $\sim \Lambda^{-2}$ is not yet resolved at $\Lambda = 16$, and the normalization of the discrete mass against $d^*\lambda$ is not established.

Status and reproducibility

Calibration is exact to the published digits; the semi-local objects are computed with exact cell averaging and covered by fourteen recomputing tests (`tests/test_semilocal_*.py`, suite of 128). The sign reversal of the conditioned remainder is a measurement on twenty test functions at two window lengths (a Rayleigh quotient scan, not a spectrum), not a theorem; the identification of the surplus with the $\log \Lambda$ term of 1999 is an inference, supported by the log-divergence of δ_S at 1. What the note establishes is negative and precise: the first semi-local step of the Connes–Consani program cannot be taken by transporting their $\Lambda = 1$ Sonin mechanism; it needs a term subtracted before the compact part can be examined. Nothing here bears on RH.